2.1. Notes

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1. Learning outcomes

By the end of this lesson, you should be able to:

- Examine 5G concepts and architecture.

Prescribed Reading

- (Nath etal, 2023)

Chapter 1

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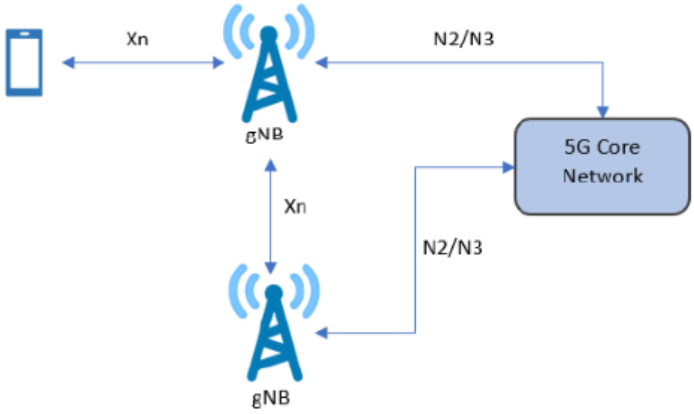
2. 5G NR and NG-RAN

- **5G NR** stands for 5G New Radio, which is the global standard for the air interface of 5G networks.
- It encompasses the radio access technologies and frequencies that enable the communication between devices and the cellular network.
- 5G NR is a key component of the broader 5G ecosystem, providing high data rates, lower latency, and improved connectivity for a wide range of applications.

- **NG-RAN** stands for Next-Generation Radio Access Network. It refers to the evolved radio access network that supports 5G NR.
- NG-RAN includes the base stations (gNBs - gNodeBs) and other components that facilitate wireless communication between user devices and the core network in 5G.
- NG-RAN plays a crucial role in delivering the enhanced performance and capabilities promised by 5G technology.

2.1. NG RAN Architecture

The Next-Generation Radio Access Network (NG-RAN) architecture is a critical component of 5G networks, designed to provide enhanced performance, flexibility, and support for various use cases. The diagram below show a high level architecture of the architecture .



The following are the key elements of the architecture.

- **gNB (Next-Generation NodeB):**
    - The gNB is the central component of the NG-RAN architecture. It serves as the base station in 5G networks and is responsible for radio communication with user devices (UEs).
    - gNBs are designed to support various frequency bands and deployment scenarios, including both standalone (SA) and non-standalone (NSA) modes.
- **Central Unit (CU):**

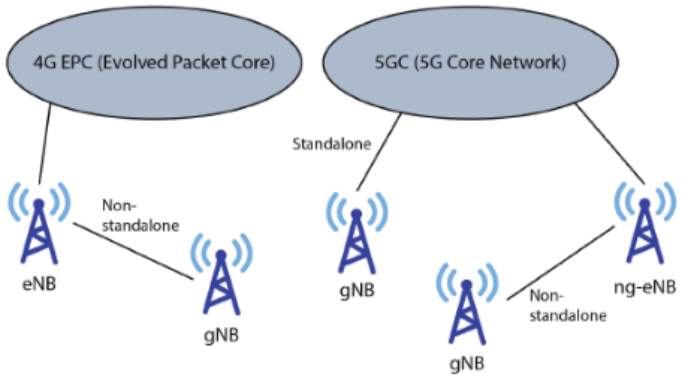
    - The CU is responsible for the non-real-time processing of data and control plane functionalities. It provides functions such as radio resource management, connection management, and mobility management.
    - The CU is further divided into the Centralized Unit (CU-CP) and the Distributed Unit (CU-DU) for increased flexibility and scalability.
- **Distributed Unit (DU):**
    - The DU is responsible for real-time processing of the data plane functionalities. It handles tasks such as radio modulation/demodulation and signal processing.
    - The DU can be physically distributed across the network, allowing for better resource utilization and reduced latency.
- **NG-RAN Architecture Options:**
    - NG-RAN supports various deployment options, including Cloud RAN (C-RAN), where the processing functions are virtualized and centralized in cloud data centers, and Distributed RAN (D-RAN), where functions are distributed at the cell site.
- **Functional Split Options:**
    - The architecture allows for different functional split options between the CU and DU, providing flexibility in how processing tasks are distributed. Functional split options include the Option 2 (where CU and DU are separated) and Option 7 (where CU and DU are combined).
- **X2 Interface:**
    - The X2 interface facilitates communication and coordination between different gNBs in the same NG-RAN, enabling features like handovers and interference coordination.
- **Fronthaul:**
    - The fronthaul is the network segment that connects the gNB with the CU and DU. It plays a crucial role in ensuring low-latency and high-bandwidth connectivity between these components.

2.2. 5G RAN deployment Options

There are some strategy options for service providers in terms of migration. There are two deployments for 5G RAN which are:

- 4G EPC (Evolved Packet Core)
- 5GC (5G Core Network)

The deployment options are illustrated using the diagram below.



2.3. Evolved Packet Core (EPC)

Evolved Packet Core (EPC)

The Evolved Packet Core (EPC) is a crucial component in the architecture of Long-Term Evolution (LTE) and, by extension, 5G mobile networks. It serves as the core network that manages and controls the flow of data between user devices, such as smartphones or IoT devices, and the external networks.

The EPC is designed to handle various types of traffic and services, ranging from traditional voice calls and SMS to high-speed data transfers and emerging applications in the Internet of Things (IoT). As 5G networks evolve, some elements of the EPC may be further enhanced or replaced to meet the specific requirements of the next generation of mobile communication.

The EPC comprises several key elements , each playing a specific role in supporting the functionalities and services provided by the mobile network:

1. **Mobility Management Entity (MME):**
    - The MME is responsible for managing the mobility of user devices within the network. It handles tasks such as tracking device location, authentication, and handovers between different base stations.
2. **Serving Gateway (SGW):**
    - The SGW acts as a gateway that connects the EPC to the radio access network (RAN). It manages the data traffic between the user device and the external networks, optimizing the routing of data packets.
3. **Packet Data Network Gateway (PDN-GW):**
    - The PDN-GW provides connectivity between the mobile network and external packet data networks, such as the internet or private networks. It is responsible for assigning IP addresses to user devices and managing data sessions.

4. **Home Subscriber Server (HSS):**
    - The HSS stores subscriber information, including user profiles, authentication credentials, and subscription details. It plays a crucial role in the authentication and authorization of users within the network.
5. **Policy and Charging Rules Function (PCRF):**
    - The PCRF is responsible for managing policy control and charging functions within the EPC. It determines the policies related to Quality of Service (QoS), bandwidth allocation, and charging for the services provided.
6. **Unified Data Management (UDM):**
    - In 5G networks, the UDM replaces the HSS and provides enhanced subscriber data management. It stores user-related information and supports authentication and authorization procedures.
7. **Authentication Server Function (AUSF):**
    - In 5G networks, the AUSF is responsible for authenticating and authorizing users. It works in conjunction with the UDM to ensure secure access to the network.

2.4. 5G Core Network

- The 5G Core Network, often referred to as the 5GC, is a critical element of 5G architecture that provides the central intelligence and management for 5G mobile networks.
- It introduces new capabilities and features compared to its predecessors, playing a fundamental role in delivering the promises of 5G technology.
- The 5G Core Network is designed to support a variety of services, including enhanced mobile broadband (eMBB), massive machine-type communication (mMTC), and ultra-reliable low-latency communication (URLLC).
- Its architecture allows for flexibility, scalability, and efficient handling of diverse use cases in the 5G ecosystem.

The following are the key elements of 5GC.

1. **Service Management and Control (SMC):**
    - The SMC oversees the lifecycle management of network services, including their creation, modification, and termination. It is responsible for service orchestration and control.
2. **User Plane Function (UPF):**
    - The UPF is responsible for handling the user data plane traffic. It manages the packet routing, forwarding, and user data optimization, contributing to low latency and high data throughput.
3. **Session Management Function (SMF):**
    - The SMF manages the establishment, modification, and termination of user sessions. It handles functions such as session continuity and ensures that the user's connectivity remains seamless.
4. **Authentication Server Function (AUSF):**

- The AUSF is responsible for authenticating and authorizing users within the 5G network. It ensures secure access to services and data.

5. **Access and Mobility Management Function (AMF):**
    - The AMF is in charge of managing the mobility of user devices within the 5G network. It handles tasks such as user registration, handovers, and mobility-related signaling.
6. **Unified Data Management (UDM):**
    - The UDM is responsible for storing and managing subscriber data. It replaces the Home Subscriber Server (HSS) in earlier networks and supports features like authentication and user profile management.
7. **Policy Control Function (PCF):**
    - The PCF manages policy control within the 5G network. It determines the policies related to Quality of Service (QoS), network slicing, and resource allocation.
8. **Application Function (AF):**
    - The AF allows external applications to communicate with the 5G network. It enables the creation of innovative and specialized services by interacting with the network's capabilities.
9. **Network Exposure Function (NEF):**
    - The NEF provides external entities and third-party applications-controlled access to network data. It facilitates the development of new services and applications by exposing network functionalities.
10. **Policy Charging and Rules Function (PCRF):**
    - Similar to its role in earlier networks, the PCRF manages policy control and charging functions. It plays a crucial role in determining policies related to charging, billing, and resource allocation.

3. NR and NG-RAN features

To meet the requirements of IMT-2020, such as coverage, capacity, and data rates, there are some techniques and technologies that are used in 5G Networks. We will discuss them in this section.

3.1. Dual Connectivity

Dual Connectivity in 5G networks refers to the capability of a User Equipment (UE) to simultaneously connect to two different Radio Access Technologies (RATs) or two different cells within the same RAT. This technology is designed to enhance the overall performance and reliability of the connection.

Dual connectivity is an important feature in the evolution of 5G networks, contributing to the realization of the envisioned capabilities, including

higher data rates, lower latency, and improved connectivity for a wide range of applications and services.

There are two main types of dual connectivity in 5G:

1. LTE-NR Dual Connectivity (EN-DC):

- This form of dual connectivity involves a combination of LTE (4G) and NR (5G) radio access technologies. The UE establishes a primary connection with an LTE eNB (Evolved NodeB) and a secondary connection with a 5G NR gNB (Next-Generation NodeB).

2. NR-NR Dual Connectivity (N-DC):

- In NR-NR dual connectivity, the UE simultaneously connects to two 5G NR cells, each belonging to a different gNB. This is designed to provide additional benefits such as increased data rates, improved coverage, and load balancing between the two cells

3.2. Small Cells

- Small cells are low-powered, short-range wireless communication devices that enhance network capacity and coverage, particularly in areas with high user demand. They play a crucial role in cellular network densification, supporting the deployment of 5G and improving overall network performance.
- Small cells are essential for improving network capacity, coverage, and performance, making them a critical component of modern wireless communication networks, especially in the context of 5G deployment.
- Other features are; Increased spectrum, Radio enhancement, Beam forming, beam steering,

4. Cloud Radio Access Network (Cloud RAN or C-RAN)

Cloud Radio Access Network (Cloud RAN or C-RAN) is an innovative and centralized approach to designing and deploying radio access networks in mobile communication systems. It leverages cloud computing concepts and virtualization to enhance network efficiency, flexibility, and scalability.

4.1. Key aspects of Cloud RAN

1. **Centralized Processing:**
    - In a traditional RAN, baseband processing functions are located at each cell site. In Cloud RAN, these functions are centralized in a cloud data center, allowing for more efficient resource utilization and maintenance.
2. **Functional Split:**
    - Cloud RAN introduces a functional split between the Centralized Unit (CU) and the Distributed Unit (DU). The CU handles non-real-time processing functions, such as radio resource management and connection management. The DU manages real-time processing tasks, including radio modulation/demodulation and signal processing.
3. **Flexibility and Scalability:**
    - Cloud RAN provides flexibility in deploying and managing radio access functions. It enables operators to scale resources based on demand, reducing the need for hardware upgrades at individual cell sites.
4. **Network Virtualization:**
    - Cloud RAN leverages virtualization technologies to create virtual instances of baseband processing functions. This enables operators to run multiple virtualized instances on shared hardware, leading to better resource utilization and cost savings.
5. **Interference Management:**
    - Centralized processing in Cloud RAN allows for more effective interference management. Coordinated scheduling and resource allocation can be performed centrally to optimize the performance of the entire network.
6. **Network Cost Savings:**
    - By centralizing processing functions and virtualizing network components, Cloud RAN can lead to cost savings in terms of infrastructure, maintenance, and energy consumption.
7. **Network Slicing:**
    - Cloud RAN supports network slicing, a key feature of 5G networks. Network slicing allows operators to create virtual networks with specific characteristics tailored to different use cases, ensuring optimal resource allocation for diverse services.
8. **Latency Reduction:**
    - Centralized processing can contribute to latency reduction in the network. This is particularly important for applications that require low-latency communication, such as augmented reality (AR) and virtual reality (VR).
9. **5G Evolution:**

- Cloud RAN is well-suited for the evolution of 5G networks. It supports the deployment of 5G features like massive MIMO (Multiple Input, Multiple Output), beamforming, and network slicing.

10. **Coordinated Multi-Point (CoMP) Transmission:**
    - Cloud RAN enables CoMP transmission, allowing multiple transmission points to cooperate in serving a user, enhancing coverage and throughput.

4.2. Activity

Activity

Discuss the benefits of implementing Cloud RAN